PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Ghalib Farras Ahmad
Program/Batch	Cybersecurity Program | NetworkWalks
Date	11 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. networkwalks.com (assigned training scope) 2. Local lab network: 10.0.0.0/24
Permission secured from client?	Training-scope activity and own lab network; authorization document not included in supplied files
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Host Discovery & Network Topology

1. Liability Disclaimer

This report records reconnaissance and local network discovery activities completed within the assigned Week 2 training scope. Testing was limited to the domain information-gathering exercise and the local network environment identified in this report. No exploitation, credential testing, or attempts to access protected resources were conducted. The report is intended for educational review and must not be treated as evidence of a confirmed security breach.

2. Introduction

This report documents two practical activities: footprinting of networkwalks.com using the assigned Kali Linux toolset (W2-PM1), followed by host discovery and topology review on the local 10.0.0.0/24 network using Zenmap (W2-PM5). The activities provide two complementary views: public information that a domain exposes and the devices visible within a permitted local network.
The evidence supplied for this report includes WHOIS, WhatWeb, nslookup, cURL, WAFW00F and a Zenmap topology export. DNSRecon remains listed among the assigned tools, but no DNSRecon output or screenshot was included in the evidence package; therefore, this report does not attribute any specific DNSRecon finding to the target. Each interpretation below distinguishes a tool observation from a security weakness that would require separate validation.

3. Tools Used

The following tools were included in the practical activities and are described according to their role.

Tool	Purpose
Kali Linux & Windows	Operating environments used for domain reconnaissance and local network review.
WHOIS	Retrieves public domain registration, registrar and name-server information.
WhatWeb	Identifies web server, CMS, plugin and client-side technology fingerprints.
nslookup	Queries DNS resolvers to obtain domain-name resolution results.
cURL -I	Requests HTTP response headers without downloading the full page body.
WAFW00F	Checks whether a web application firewall can be fingerprinted from its responses.
DNSRecon	Enumerates DNS records; no raw result or screenshot was included in the supplied evidence.
Zenmap (Nmap GUI)	Runs Nmap scans and visualizes discovered hosts and network topology.
Windows CMD	Used in the lab workflow to inspect local IP and network configuration.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I examined networkwalks.com through a sequence of focused information-gathering checks. Rather than treating every returned value as a vulnerability, I considered what each result reveals, how reliable the observation is, and what additional validation would be needed before raising a security finding.

WHOIS identifies GoDaddy.com, LLC as registrar and lists NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM as name servers. The captured record shows creation on 6 November 2019 and expiry on 6 November 2027. These details describe registration and DNS delegation, not a security weakness.

WhatWeb reported Apache, WordPress 7.1.4, WordPress Download Manager 3.3.76, jQuery 3.7.1 and Bootstrap 7.1.4. The output also contains an execution-expired message, so the fingerprints should be treated as indicative and verified against the owner's software inventory.

nslookup queried Google Public DNS (8.8.8.8) and returned 192.232.216.135 for networkwalks.com. This is the answer observed at capture time; DNS responses can vary by resolver and time.

cURL received HTTP/2 200 and returned Apache, a text/html content type and a WordPress REST API link at /wp-json/. The link shows an advertised API route, not an authorization flaw. The cookie-like value has been redacted from the report image.

WAFW00F 2.4.2 reported ModSecurity (SpiderLabs) after two requests. This fingerprints a possible protective layer but does not establish its rule coverage or effectiveness.

DNSRecon was included in the toolset, but no raw output or screenshot was supplied. No DNS record or DNSRecon-derived conclusion is therefore claimed in this report.

4.2 Network Scanning with Zenmap

The Zenmap topology shows 10.0.0.1, 10.0.0.2 and 10.0.0.12. The lab configuration records 10.0.0.1 as the gateway address and 10.0.0.2 as the Kali VM address. The role of 10.0.0.12 as the Windows 10 VM Address.

The lab subnet was 10.0.0.0/24. The supplied topology supports the three visible IP addresses

The topology export displays these addresses:

- 10.0.0.1 — Gateway addresss
- 10.0.0.2 — Kali Linux VM address
- 10.0.0.12 — Windows 10 VM address

The diagram confirms the nodes shown in Zenmap's topology view; it does not establish their services, ownership or authorization status.

For a complete inventory, retain the original Zenmap scan output and verify each address against the router, DHCP lease table or authorized asset inventory.

5. Risk Analysis / Impact

The following table summarizes the observations that can be supported by the supplied evidence. Risk ratings describe information exposure or follow-up priority, not confirmed exploitable flaws.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology fingerprint available	WhatWeb returned Apache, WordPress 7.1.4 and plugin/library fingerprints; one execution-expired message also appeared.	Could help an observer prioritize further research if the fingerprints are accurate; version status needs owner-side verification.	● Medium
2	Domain address resolved	nslookup via 8.8.8.8 returned 192.232.216.135.	Provides a network locator for the domain, which is normal for a public website and not a vulnerability by itself.	● Informational
3	HTTP response metadata visible	cURL returned HTTP/2 200, Apache and a WordPress REST API link at /wp-json/.	May assist technology identification; the header does not demonstrate an API authorization weakness.	● Low
4	WAF product fingerprinted	WAFW00F 2.4.2 reported ModSecurity (SpiderLabs) after two requests.	Reveals a defensive technology indicator, but does not establish rule coverage or bypassability.	● Informational
5	Additional local node requires identification	Zenmap topology shows 10.0.0.12 alongside 10.0.0.1 and 10.0.0.2.	If the node is not recognized by the network owner, asset verification is appropriate; the screenshot does not prove it is unauthorized.	● Medium
6	DNSRecon result not evidenced	No DNSRecon output or screenshot was included in the evidence ZIP.	No technical impact can be inferred; the report cannot substantiate DNSRecon-derived records.	● Informational

Risk level key: ● Medium ● Low ● Informational

The observations relate to information disclosure and network visibility. None of the supplied screenshots demonstrates successful exploitation, unauthorized access, or a confirmed vulnerability.

The practical work focused on reconnaissance and host discovery. No port or vulnerability scan, credential attack, exploitation, persistence, or post-exploitation action is documented.

A technology fingerprint, resolved IP address, HTTP header or visible host is not automatically a security defect. A finding should only be upgraded after the relevant owner confirms the asset and a controlled test demonstrates a concrete security impact.

6. Recommendations

The following actions would improve the accuracy of the inventory and reduce unnecessary exposure:

1. Verify web technology versions
   Compare WhatWeb fingerprints with the owner's software inventory before raising patch-related findings.
2. Review response metadata
   Limit unnecessary server and application details in HTTP responses where this can be done without affecting service behaviour.
3. Maintain DNS records
   Confirm public records remain current and map only to services intended for external access.
4. Check API authorization
   Review authentication and object-level authorization for relevant WordPress REST API resources.
5. Review WAF controls
   Validate ModSecurity coverage, rules and logging with the system owner; fingerprinting alone does not prove effectiveness.
6. Identify 10.0.0.12
   Confirm the device through an authorized asset inventory or network console.
7. Keep scan evidence
   Save Zenmap output, command, timestamp, subnet and topology together.
8. Retain DNSRecon output
   Add the raw output and screenshot before making DNSRecon-specific claims.
9. Respect authorization
   Repeat active testing only within the approved scope.

7. Conclusion

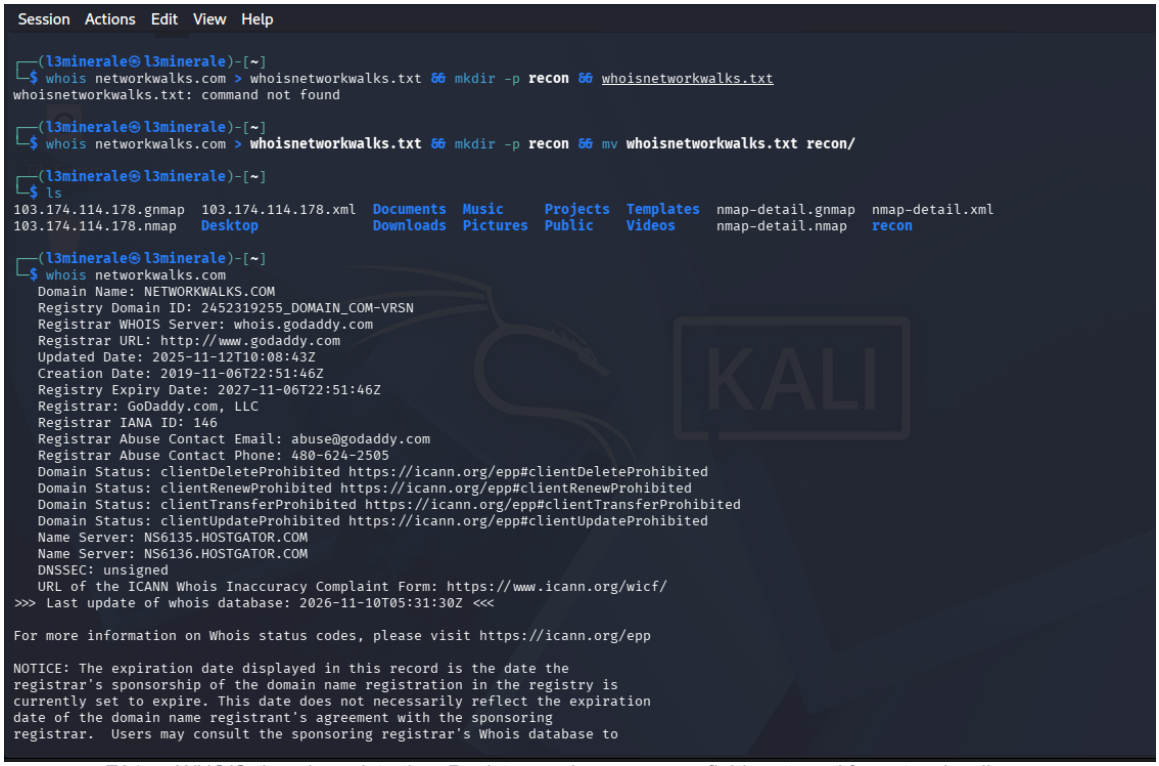
The Week 2 exercises covered domain footprinting and local network discovery. They show how public metadata and local host visibility contribute to an initial environment assessment.

WHOIS, WhatWeb, nslookup, cURL and WAFW00F produced the observations described above. DNSRecon remained in the toolset, but its output was not supplied and is not interpreted.

The Zenmap topology displayed 10.0.0.1, 10.0.0.2 and 10.0.0.12. The first two align with the lab host and Kali VM configuration; and Windows VM

The exercise reinforced that reconnaissance produces leads, not verdicts; each finding needs context and validation.

8. Evidences Collected



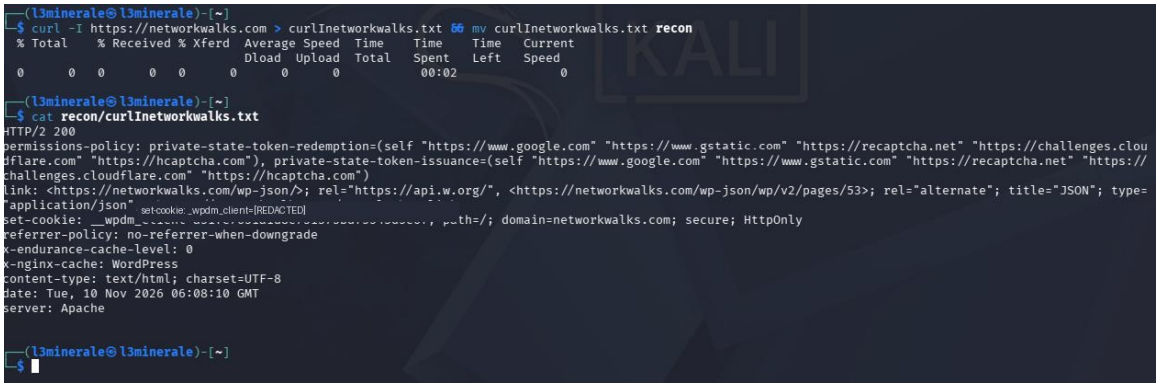
E01 — WHOIS domain registration. Registrar and name-server fields returned for networkwalks.com.



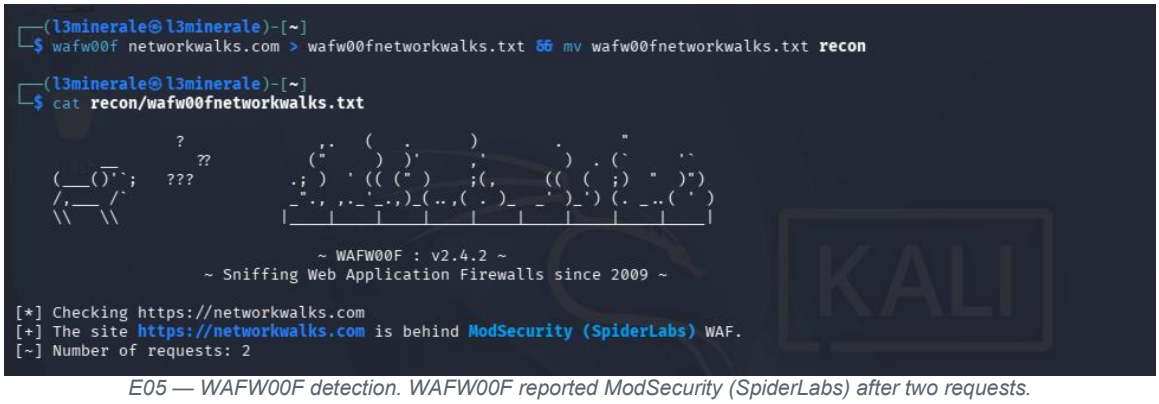
E02 — WhatWeb technology fingerprint. Web technology indicators reported by WhatWeb; versions remain unverified fingerprints.



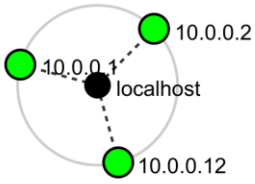
E03 — DNS resolution with nslookup. Google Public DNS returned 192.232.216.135 for networkwalks.com.



E04 — HTTP response headers with cURL. HTTP/2 200 response and selected headers; cookie value redacted.



E05 — WAFW00F detection. WAFW00F reported ModSecurity (SpiderLabs) after two requests.



E06 — Zenmap network topology. Topology export showing 10.0.0.1, 10.0.0.2 and 10.0.0.12.

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📌 Project Information
Program Name: Cybersecurity Program at NetworkWalks | Week: 02 | Repository: NETWORKWALKS-GHALIB-B084E-WK2-CYBERSECURITY-RECON